IQOO HACKATHON 2026 · GRAND FINALE · OPEN INNOVATION

थकान

Thakaan

Garmi se pehle, Thakaan.

A phone-only, offline heat-strain guardian for outdoor workers, with Chhaya Route to steer them to shade and water before they collapse.

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THE PROBLEM

Heat takes judgement before it takes the body.

Noon, 41 °C. A delivery rider has been out for hours. He will not feel heat illness coming, and the phone is already in his hand.

No warning

Heat dulls judgement first, so workers push on until they collapse.

Rest costs money

Paid per task: every break is lost income.

No wearable

They carry a phone all day but cannot afford a heat monitor.

Alerts aren't personal

City alerts say "it is hot today", nothing about you.

THE RED-LIGHT SPIKE

On the bike, the wind cools. Stop over hot tarmac and the airflow vanishes; the body heats in seconds. Tools built for soldiers and labourers miss this.

THE IDEA

Turn the phone they already carry into a heat-strain guardian.

One phone. No wearable.

Fully offline

Body + brain + sun → one level → one action

HOW IT WORKS

Three watchers, one score, one action

Body
Fingertip pulse at stops → core-temperature trend

Brain
Tremor, tapping, speech, symptoms vs your own normal

Sun
Light sensor, satellite sky view, cached weather → WBGT

STRAIN SCORE

0–100

Every minute, with a 15 and 30-minute forecast. Never blank: the sun layer is always there.

Spoken alert
One action, in the worker's language, with its time cost

Chhaya Route
Shade and water detour sized to current strain

Supervisor view
Levels only, never raw heart rate or voice

SENSORS

11 sensors, all inside the phone

Accelerometer · 200 Hz
Activity, stops, falls, tremor, gait, chest heartbeat

Gyroscope · 200 Hz
Lifting vs walking; stillness for the chest heartbeat

Light sensor · ~5 Hz
Sun or shade right now; sunlight strength in W/m²

Colour-temp sensor
Blue shade light vs yellow direct sun

GPS position · 1 Hz
Map position, routing, moving or stopped, riding wind

GPS satellites · 1 Hz
Sky view: weak satellites near the sun mean a tree or roof overhead

Rear camera + torch · 30 fps
Fingertip heart rate and HRV; street photo for sky share

Front camera · 30 fps
Face pulse, breathing, eye closing, yawns

Microphone
Spoken digit test; "pani piya" drink logging

Touchscreen
Tapping speed and accuracy, reaction time

Phone temperature
Blocks checks when the phone is too hot; nudges air temperature

+ Saved offline
Weather and PM2.5 forecast, OSM road graph with shade layer

WHAT THE WORKER SEES

Four levels, never a temperature

Cool

Score 0–29

Fastest route.
Passive logging.

Warming

Score 30–54

Drink nudge. Small shade detours allowed.

"Paani pee lo, agle signal pe."

Strained

Score 55–79

Water stop routed next. Full check at next stop.

"3 minute aage chhaya aur paani hai."

Danger

Score 80–100

Stop now. Nearest shade, contact and supervisor alerted, cooling coach.

"Abhi ruk jao. Chhaya mein baitho."

Anchors force a level up on one alarming fact, even if the blended score looks calm.

No flicker. Danger clears only after a recovery check. Every change logs its top three reasons.

WATCHER 1 · BODY

A hidden core-temperature trend, from the heartbeat

01

Fingertip pulse

Finger on camera and torch at stops. Camera locked at a steady 30 fps. Bad captures are rejected with a reason, never guessed.

02

ECTemp Kalman filter

Published equations reproduced exactly: 7 of 7 reference vectors pass on the phone. Gaps grow uncertainty, never shrink it.

03

Heat-balance prior

Gagge two-node and ISO 7933 sweat models predict warming from sun, effort and clothing; drinks are logged.

≤ 38.5 °C

Every core-trend action fires at or below 38.5 °C (38.0 °C before acclimatisation), well before the danger zone. A separate rule raises the level on a sustained resting heart rate above 90–100 bpm.

WATCHER 2 · BRAIN

Catching clumsiness before the worker feels it

A ~90-second check at a hand-over

10–15 s	**Tremor hold**: 4–12 Hz shake while holding still
20 s	**Fitts tapping**: hand-eye throughput and miss rate
5 s	**Speech code**: say digits, on-device Whisper-tiny
2 taps	**Symptoms**: thirst, headache, dizziness, cramps

Your own normal

Compared only with the same worker, same phone, same time of day. A CUSUM "bucket" fills with small repeated slowdowns.

Independent evidence

Two different families (motion, touch, voice, navigation, symptoms) plus high heat force at least Strained.

WATCHER 3 · SUN

How much heat is landing on you right now

Sun or shade

Light sensor: brightness drop plus colour shift (sun ~5,000 K, shade ~7,500–10,000 K)

What blocks the sky

Per-satellite line-of-sight model tells a roof, a flyover and a tree apart

Air

Cached forecast for temperature and humidity: works offline

Breeze

Bike speed is airflow. At a red light it drops to zero

→ ONE HEAT-STRESS NUMBER

Liljegren WBGT via the **lwbgt** C library on the NDK. Always available, so the strain score never goes blank.

WHEN UNSURE, ASSUME WORSE

Lost satellites count as sun. Missing weather counts as hot.

CHHAYA ROUTE · छाया

The fastest road for how your body is doing right now

COST PER ROAD SEGMENT

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time × (1 + λ(strain) × sun)
+ heat-island penalty
```

λ = 0 below score 30, rising to 3 by score 75. Strain is the dial; λ is the wire to the map.

Own Dijkstra on an OpenStreetMap graph. Shade ray-cast from building and canopy heights into an H3 grid, 4 seasons × 30-minute slots. Rider flood reports close roads for 3 hours.

Cool	Fastest route, no detour
Warming	Up to +10% time for shade
Strained	Up to +25%, plus a water stop
Danger	Nearest cool seated stop

Every route shows its price: extra seconds and % shade.

CHHAYA ROUTE · MAP LAYERS

How the map knows shade, smoke and water

Shade

Sun position ray-cast against building and tree-canopy heights, baked into a ~25 m hex grid for every 30 minutes across 4 seasons. Riders' light and satellite readings correct it.

Pollution

Cached PM2.5 on the CPCB scale. Combined with breathing rate from heart rate into a shift air dose.

Flooding

One-tap rider report with a fresh photo closes every road within 30 m for 3 hours. Rain onset is a cue to check.

Traffic and signal

Typical rush-hour traffic plus the rider's own measured speed; phone signal logged from the phone's own radio.

OPTIONAL · CREW MESH

Riders' phones pass live updates to each other

Peer-to-peer over Bluetooth LE, no internet needed. Flood reports spread along the road; a buddy's phone buzzes if a teammate hits Danger.

3 hops · Anonymous · Rate-limited

BUILT FOR THE IQOO 15

Everything runs on the phone, with no network

Job	Method	Runs on
Body filter, pulse DSP, routing, WBGT	Kotlin + NDK, lwbgt in C	CPU
Activity, chest pulse, forecast, sky view	LiteRT small models	NPU / CPU
Speech code (en, hi, kn)	Whisper-tiny, Vosk	On-device
Alert wording, cooling coach	Gemma 4 E2B, LiteRT-LM	Adreno GPU
Voice out: en, hi, kn, ta, te	Android offline TTS	CPU

LLM GUARDRAIL

Gemma only re-words the engine's line. Any line that adds a number, quotes a temperature or reassures is rejected and the template is used. The level never comes from model text.

MODELS

5 models we made, 5 we put on the phone

MADE BY US

activity_cnn · 1D CNN
2 s of motion → idle, riding, walking, stairs, lifting

scg_denoiser · dilated 1D CNN
Cleans the chest heartbeat vibration, 5 s at 100 Hz

POS · colour-maths pulse
Pulse from fingertip and face skin colour

strain_gru · GRU, 24 units
Last 20 min of strain and sun → score at +15 and +30 min

los_xgboost · XGBoost, 40 trees
Per satellite: clear sky, or blocked by a tree or building

CONVERTED TO RUN ON THE PHONE

Gemma 4 E2B · 1.9 GB, GPU
Rewords the alert sentence, behind a guard

MediaPipe Face Landmarker
478 face points: eye closing and yawns

EfficientPhys
Learned pulse from 72×72 face frames

SegFormer-B0
Sky, tree or building, pixel by pixel

Vosk · English-India + Hindi
Offline digit recogniser for the speech test

MATHS MODELS

ECTemp core trend · Gagge heat balance · Liljegren WBGT · NOAA sun · CUSUM. These set the level; AI gives a second opinion or rewords.

PRIVACY

The worker owns the data

Data	Leaves the phone?
Camera frames, audio	Never (RAM only)
Heart rate, core trend, brain scores, symptoms	Never
Strain level and alerts	Only to people the worker approves
Light and sky readings	Anonymised, snapped to map cells
Exposure log	Only when exported

No INTERNET permission

Verified on the iQOO 15 with dumpsys.

Supervisor sees levels

Level, trend and coarse reasons.
Never raw heart rate or voice.

THE VISION

From an app to a heat-safety standard for every gig worker

01 · Platforms

Built into delivery apps

A heat-safety layer every rider gets on the phone they already use.

02 · Compliance

Proof of safe shifts

The signed exposure log shows rest, water and shade breaks were offered and taken. Levels only, never health data.

03 · Government

A mandated safeguard

Heat-strain monitoring as a compliance requirement for platforms under gig-worker welfare rules.

No gig worker should die of heat on the job.